

Miguel Aréjula Aísa

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Profile

Software Engineering Master's student at SDU and former MongoDB intern with a focus on system and software architecture. Experienced in designing end-to-end web applications, scalable backends, and cloud/database architectures using TypeScript, Python, and modern web frameworks. Passionate about turning complex requirements into clean, modular designs and production-ready systems.

Experience

Industry Solutions Intern, MongoDB

July 2026 - September 2026

- Built an offline-first wallet demo for Financial Services: an on-device ObjectBox store synced bidirectionally with MongoDB Atlas, payments queued offline and replayed on reconnect, semantic transaction search on Voyage AI embeddings, and a LangGraph assistant on a local Ollama model.
- Maintained and extended the Next.js demo-deployment tool used across the Industry Solutions team, shipping bug fixes and new features to production.
- Wrote a Claude Code skill that drafts a full MongoDB Solutions Library article from a solution repository, encoding the team's template, taxonomy, and style rules so every write-up starts consistent instead of from a blank page.
- Shared the solution externally through a blog post, a MongoDB Solutions Library entry, and a YouTube video.

Teaching Assistant, SDU

February 2026 - May 2026

- Supported students in understanding core software architecture concepts by assisting with lab sessions, assignments, and problem-solving, while clearly explaining technical concepts and collaborating with professors.

Software Engineer, University of Zaragoza

June 2024 – August 2025

- Led end-to-end development of a clinical web app for Hospital Clínico Lozano Blesa, covering medical stakeholder requirements gathering, UI prototyping, and system delivery.
- Developed a full-stack system using React and PostgreSQL, with RESTful APIs in Express and FastAPI, streamlining operations for ~50 patients/day.
- Integrated structured clinical data into a relational model, validating clinical pathways through Petri nets and formal methods.

Projects

Leafy Wallet

github.com/leafy-wallet

- Offline-first personal wallet demo for Financial Services: an on-device ObjectBox store syncing bidirectionally with MongoDB Atlas, with payments queued offline and replayed on reconnect.
- Implemented multi-document ACID transactions, reconciliation against an external payment platform (PSP) behind an OAuth PKCE SSO flow, and semantic transaction search on Voyage AI embeddings (Atlas \$rankFusion online, on-device HNSW offline).
- Built a LangGraph assistant on a local Ollama model, routing questions to MongoDB MCP tools online and to the on-device store offline. **Tools Used:** Next.js, FastAPI, Python, MongoDB Atlas, ObjectBox, LangGraph, Ollama, Docker, Figma.

AgroNet

github.com/AgroNet

- Designed and developed a full-stack collaborative web platform for farmers, integrating real-time market prices, personalized weather alerts, and interactive geospatial data visualization across Spain.
- Managed communication with the backend hosted on AWS, ensuring reliable API integration and real-time data synchronization. **Tools Used:** Astro, Node.js, TypeScript, MongoDB, AWS, Vercel, Figma.

Energy Price Prediction

github.com/BigDataProject

- Using Danish Meteorological Institute (DMI) weather datasets and national energy consumption/price data to analyze correlations and forecast future electricity prices.
- Expected insights include: renewable energy impact, peak price periods, seasonal patterns, and cost optimization windows.
- Applying big data frameworks and machine learning models to process large-scale datasets and deliver predictive analytics. **Tools Used:** Kafka, HDFS, Kubernetes, Spark, Python, Hive

Skills & Tools

Languages & Core: TypeScript, JavaScript, Python, Go, C/C++, Java, SQL, HTML, CSS.

Web Development: React, Next.js, Astro, Node.js, Express, FastAPI, Tailwind CSS, Figma, REST APIs.

AI & Agentic Workflows: LangGraph, Tool Calling, MCP Tools, Claude Code Skills, LLM Integration (Ollama), Vector Search.

Infrastructure & Tools: PostgreSQL, MongoDB Atlas, Docker, AWS, Vercel, Git.

Education

Master in Software Engineering, University of Southern Denmark

Sept 2025 – June 2027

Bachelor in Software Engineering, University of Zaragoza

Sept 2021 – June 2025

- Achieved high honors in Information Systems II and Verification and Validation, demonstrating exceptional proficiency and understanding in these subjects.